

A Superexponentially Weighted Hilbert Space with Only Constant Multipliers

Candidate solution to Question 4.9 of arXiv:2008.03700

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Abstract

We give an affirmative answer to Bilokopytov's Question 4.9. The Hilbert space of power series with coefficient norm

$$\|f\|_{\mathcal{H}}^2 = \sum_{n \geq 0} |a_n|^2 e^{2n^2}, \quad f(z) = \sum_{n \geq 0} a_n z^n,$$

consists in fact of entire functions, so it is a Hilbert space of holomorphic functions over the unit disk. If a multiplier has a nonzero Taylor coefficient of degree $j \geq 1$, its action on normalized monomials has norm at least $|b_j| e^{2jn+j^2}$, which is unbounded as $n \rightarrow \infty$. Consequently the multiplier algebra is exactly $\mathbb{C}$.

Source Problem

The source is Eugene Bilokopytov, *Multiplier algebras of normed spaces of continuous functions*, arXiv:2008.03700 [1]. The open problem occurs as Question 4.9 on page 13 of the arXiv PDF, immediately after Corollary 4.8.

Let us show that $\mathbf{H}$ is an algebra. Since $\mathcal{H}^2 \subset \mathcal{H}^1$ it follows that $\mathbf{H} \subset A(\mathbb{D}) \subset \mathcal{H}^\infty$ (see [17] Theorem 3.11). For $f, g \in \mathbf{H}$ we have $(fg)' = fg' + f'g$. Since $f, g \in \mathcal{H}^\infty = \text{Mult}(\mathcal{H}^2)$ it follows that $fg', f'g \in \mathcal{H}^2$, and so $fg \in \mathbf{H}$. Thus, from Proposition 2.9 $\mathbf{H} = \text{Mult}(\mathbf{H})$ (as topological vector spaces). $\square$

We conclude the section with some remarks regarding the multipliers of NSHF's. First, similarly to the continuous case, if X is a domain in $\mathbb{C}^n$, $\mathbf{F}$ is a 1-independent NSHF over X , and $\mathbf{E}$ is a NSHF over X , then $\text{Mult}(\mathbf{F}, \mathbf{E}) \subset \mathcal{H}(X)$. For the multiplier algebras an even stronger fact is true (see [7] Proposition 4.3).

Proposition 4.7. *Assume that X is a domain in $\mathbb{C}^n$, and let $\mathbf{F}$ be a NSHF over X . Then, $\text{Mult}(\mathbf{F})$ contractively embeds into $\mathcal{H}_\infty(X)$ in the sense that if $T \in \mathcal{L}(\mathbf{F})$ is a MO, there is a unique holomorphic $\omega : X \rightarrow \mathbb{C}$ such that $T = M_\omega$, for which also $\|\omega\|_\infty \leq \|M_\omega\|$.*

Since $\mathcal{H}_\infty(\mathbb{C}^n) = \mathbb{C}$, applying Proposition 3.7 to the monomials we get the following result.

Corollary 4.8. *Every NSHF over $\mathbb{C}^n$ is anti-multiplicative. For every separable Banach space F there is an anti-multiplicative 1-independent BSCF isometrically isomorphic to F .*

Note that our examples of anti-multiplicative NSCF's exploit some kind of pathologies either of the phase space, or the normed space. It is natural to ask if one can find an example where both are as well-behaved as possible.

Question 4.9. *Does there exist an anti-multiplicative HSHF over $\mathbb{D}$?*

The question reads:

Does there exist an anti-multiplicative HSHF over $\mathbb{D}$?

Definitions

Following the source, a *Hilbert space of holomorphic functions* (HSHF) over $\mathbb{D}$ is a Hilbert space $\mathcal{H} \subseteq \mathcal{O}(\mathbb{D})$ whose norm topology is stronger than the compact-open topology. Equivalently in the Hilbert setting, all point evaluations are bounded; the closed graph/uniform boundedness argument then gives the required compact-open control.

A function $\varphi : \mathbb{D} \rightarrow \mathbb{C}$ is a *multiplier* of $\mathcal{H}$ if $M_\varphi f = \varphi f$ defines a bounded operator $M_\varphi : \mathcal{H} \rightarrow \mathcal{H}$. The space is *anti-multiplicative* when $\text{Mult}(\mathcal{H}) = \mathbb{C}\mathbf{1}$, i.e. its only multipliers are constants.

Proof Intuition

In a coefficient Hilbert space with monomial weights w_n , multiplication by z^j shifts degree n to degree $n+j$ and has size w_{n+j}/w_n on the normalized n th monomial. We choose $w_n = e^{n^2}$. Then every fixed positive shift has exploding ratio

$$\frac{w_{n+j}}{w_n} = e^{(n+j)^2 - n^2} = e^{2jn+j^2} \rightarrow \infty.$$

For an arbitrary nonconstant multiplier, isolate any one nonzero positive Taylor coefficient. Its contribution to a single output coefficient already forces exactly the same explosion; no cancellation can occur because each input is a monomial.

The only other obligation is to make the Hilbert space genuinely a function space on $\mathbb{D}$. The reciprocal weights decay faster than every geometric sequence, so Cauchy–Schwarz gives uniform convergence on every compact subset of the whole plane.

Main Result

Theorem 1. *There exists a 1-containing, infinite-dimensional HSHF $\mathcal{H}$ over $\mathbb{D}$ such that*

$$\text{Mult}(\mathcal{H}) = \mathbb{C}\mathbf{1}.$$

In particular, Question 4.9 of [1] has an affirmative answer.

Proof. Define

$$\mathcal{H} = \left\{ f(z) = \sum_{n=0}^{\infty} a_n z^n : \|f\|_{\mathcal{H}}^2 := \sum_{n=0}^{\infty} |a_n|^2 e^{2n^2} < \infty \right\}.$$

The coefficient map

$$f \mapsto (a_n e^{n^2})_{n \geq 0}$$

is an isometry onto $\ell_2(\mathbb{N}_0)$, so $\mathcal{H}$ is a Hilbert space. To see that its elements define holomorphic functions with the required topology, fix $R > 0$. Cauchy–Schwarz gives, for $|z| \leq R$,

$$\begin{aligned} |f(z)| &\leq \sum_{n \geq 0} |a_n| R^n \\ &\leq \left(\sum_{n \geq 0} |a_n|^2 e^{2n^2} \right)^{1/2} \left(\sum_{n \geq 0} R^{2n} e^{-2n^2} \right)^{1/2} \\ &= C_R \|f\|_{\mathcal{H}}, \end{aligned}$$

where $C_R < \infty$ by the ratio test. Hence the defining power series converges uniformly on every disk $|z| \leq R$; every element of $\mathcal{H}$ is entire. The displayed estimate also makes the inclusion into the compact-open topology continuous (in particular on compact subsets of $\mathbb{D}$). Thus $\mathcal{H}$ is an HSHF over $\mathbb{D}$. It contains $\mathbf{1}$, and it is infinite-dimensional because it contains all monomials.

Let φ be a multiplier of $\mathcal{H}$. Since $\mathbf{1} \in \mathcal{H}$,

$$\varphi = M_\varphi \mathbf{1} \in \mathcal{H}.$$

Write its Taylor expansion as

$$\varphi(z) = \sum_{k=0}^{\infty} b_k z^k.$$

Suppose that φ is nonconstant. Choose $j \geq 1$ with $b_j \neq 0$. For each $n \geq 0$, put

$$u_n(z) = e^{-n^2} z^n.$$

Then $\|u_n\|_{\mathcal{H}} = 1$. The coefficient of z^{n+j} in φu_n is exactly $b_j e^{-n^2}$. Since the squared Hilbert norm is a sum of nonnegative coefficient contributions,

$$\begin{aligned} \|M_\varphi u_n\|_{\mathcal{H}} &\geq |b_j| e^{-n^2} e^{(n+j)^2} \\ &= |b_j| e^{2jn+j^2} \longrightarrow \infty. \end{aligned}$$

This contradicts the boundedness of M_φ on the unit vectors u_n . Therefore every multiplier is constant. Conversely, each constant c acts as the bounded operator $cI_{\mathcal{H}}$. Hence $\text{Mult}(\mathcal{H}) = \mathbb{C}\mathbf{1}$. $\square$

Remark (Reproducing kernel). The construction may equivalently be presented as the reproducing-kernel Hilbert space with kernel

$$K(z, w) = \sum_{n=0}^{\infty} e^{-2n^2} (z\bar{w})^n.$$

The series converges for all $(z, w) \in \mathbb{C}^2$, and its associated orthonormal basis is $e^{-n^2} z^n$. This makes explicit that the example is an ordinary RKHS, not merely a formal coefficient space.

Verification Notes

The proof has four independent checks.

1. Completeness is exactly completeness of ℓ_2 under the displayed coefficient isometry.
2. The Cauchy–Schwarz estimate holds for every $R > 0$, a statement stronger than needed for an HSHF over $\mathbb{D}$.
3. The test vectors are normalized: their only coefficient has weighted magnitude $e^{-n^2} e^{n^2} = 1$.
4. There is no cancellation in the lower bound. For a fixed monomial input, the coefficient of output degree $n + j$ comes from the unique Taylor coefficient b_j ; discarding all other nonnegative squared coefficient contributions gives a valid lower bound.

No numerical or computer-assisted claim is used.

Novelty Check And Limitations

Before promotion, the run’s registry and solution, attempt, and proof-gap indexes were searched for arXiv:2008.03700, its title, and the phrases “anti-multiplicative HSHF”, “Hilbert space of holomorphic functions”, and “only constant multipliers”. No duplicate result or prior attempt was found.

A bounded web search on 9 August 2026 used the exact text of Question 4.9, the exact term “anti-multiplicative HSHF”, and close variants involving reproducing-kernel Hilbert spaces with only constant multipliers. The search located the source paper and general background results, but no separate later paper explicitly answering Question 4.9. This was not an exhaustive bibliographic or citation-graph search, so novelty remains subject to human literature review.

The result answers the existence question exactly as stated. It does not claim additional regularity of the norm under classical operators such as the shift, nor does it classify all HSHFs with trivial multiplier algebra. The rapid weights are the mechanism that deliberately make every positive-degree shift unbounded.

Human Review Recommendation

Promote for expert review as a candidate full affirmative answer. The proof is short and self-contained. A reviewer should verify the source convention for HSHF and the coefficient-isolation argument for arbitrary multiplier symbols. Under the definitions in [1], both checks are direct: the compact-open embedding is explicit, $\mathbf{1} \in \mathcal{H}$, and Taylor coefficients are unique.

References

- [1] E. Bilokopytov, *Multiplier algebras of normed spaces of continuous functions*, arXiv:2008.03700, 2020. <https://arxiv.org/abs/2008.03700>.